
Meeting #6

July 16, 2026

Link between Parkinson's and Dysarthria

TORGO → three Parkinson's Disease (PD) datasets

Training Speech Classifier on TORGO Dataset

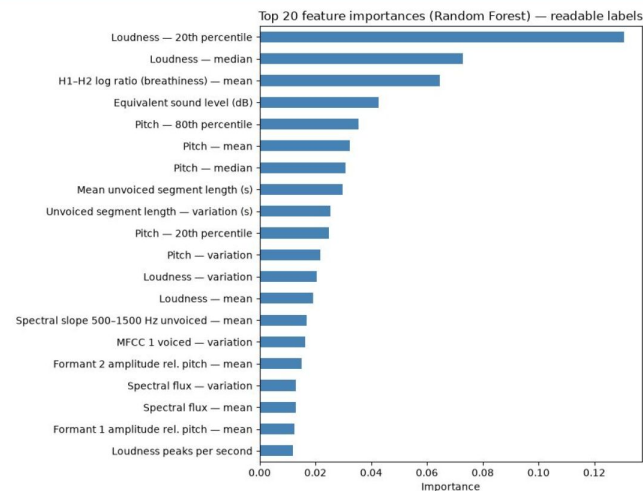
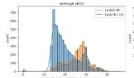
Feature Importance

- The Random Forest algorithm lets you see which features are the most important for the model's prediction!

Loudness - #1, #2, #4

Breathiness - #3

- This gives you important features overall, can use SHAP to look at features important for specific predictions / test cases
- To figure out the sign (direction of effect) of features, can use histogram feature graphs or SHAP beeswarms



Convolutional Neural Network to Model Articulation Impairments in Patients with Parkinson's Disease (2017)

J. C. Va'squez-Correa^{1,2}, J. R. Orozco-Arroyave^{1,2}, E. No'th²

The main symptoms of the impaired speech of PD patients include reduced loudness, monopitch, monoloudness, hypotonicity, breathy, hoarse voice quality, and imprecise articulation. These symptoms are typically grouped and called **hypokinetic dysarthria** [3]

Several studies in the literature have described the speech impairments of PD patients in terms of **phonation, articulation, and prosody** [4, 5, 6]. Phonation is related to the capability of the speaker to make the vocal folds vibrate to produce vocal sounds, articulation is related with the modification of the position, stress, and shape of several muscles to produce speech, and prosody reflects variation of loudness, pitch, and timing to produce natural speech.

Datasets Access

- **Bridge2AI - *In Progress***
- **Parkinson's Disease - *Downloaded***

Saarbrücken Voice Database (SVD)

- 2008
- German
- Broadly studies vocal disorders
- Only has 1 subject with "Morbus Parkinson" diagnostic label
- 687 recordings available of "healthy" speakers

NeuroVoz: a Castilian Spanish corpus

- 2024
- Spanish
- Specific to PD patients who are Castilian Spanish speakers
- 54 PD, 58 HC
- Scripted + spontaneous speech
- 3,000 audio files

Mobile Device Voice Recordings at King's College London (MDVR-KCL)

- 2017
- English
- Severity annotations in addition to diagnosis
- 16 PD, 21 HC
- Read Text + Spontaneous Dialogue
- 73 audio files



Parkinson's Disease (PD) Datasets

Copied over data onto speechlab server under
`/proj/speech/corpora`

```
[esd2154@kedi:/proj/speech/corpora/Parkinsons$ ls -la
total 61
drwxr-xr-x  5 esd2154 student  5 Jul 15 17:06 .
drwxrwxrwx 115 frank  speech 129 Jul 15 16:35 ..
drwxr-xr-x  4 esd2154 student  4 Jul 15 17:06 MDVR_KCL
drwxr-xr-x  4 esd2154 student  4 Jul 15 17:06 'Saarbrücken Voice Database'
drwxrwxrwx  7 esd2154 student  7 Jul 15 17:06 'Spanish neurovoz_v3'
```

Saarbrücken Voice Database (SVD)

Group	Files
German healthy	19,285
German Morbus Parkinson	32

NeuroVoz: a Castilian Spanish corpus

Subfolder	Files
audios/	2,976
transcriptions/	2,740
grbas/	33
metadata/	2
audio_features/	1
Total	5,753

Mobile Device Voice Recordings at King's College London (MDVR-KCL)

Subfolder	Files
ReadText/HC	21
ReadText/PD	16
SpontaneousDialogue/HC	21
SpontaneousDialogue/PD	15
Total	73

Dataset 1: German

Saarbrücken Voice Database (SVD)

Had the least # of patients (only one) for PD

Still had a lot of audio files from the HC group

Can't rly listen to this b/c nsp → wav needs to be converted (I can only play .wav files)

>  sentences

Each patient has two folders: >  vowels

They're asked to read pre-determined sentences and also all of the German vowels (a,i, aiu, u)

 remarks

"an": auch als "a" gelabelt

"uh": auch als "un", "ul" gelabelt

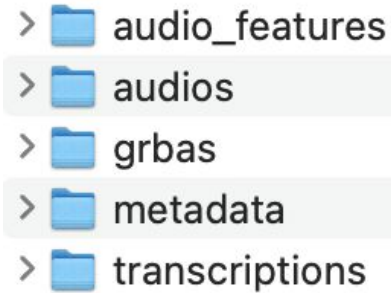
- 2008
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Dataset 2: Spanish

NeuroVoz: a Castillian Spanish corpus

Structure of the dataset:

- *audios* folder had 3,000 audio files
- Not sorted into subfolders, so the directory contains all audio files in same folder, sorted alphabetically by file name on my computer



- > audio_features
- > audios
- > grbas
- > metadata
- > transcriptions

Notes:

- Completely in Spanish
- People reading standard set of short phrases or producing certain sounds
- Name of the audio file generally describes the sound being produced (e.g. *PD_SOMBRA_0067*)
- Most of the audio files are very clear and there's no interference from moderator's speech
- However, some audio files are very quiet (audio volume varies) and are short in length (<5 sec)

- 2024
- Spanish
- Specific to PD patients who are Castillian Spanish speakers
- 54 PD, 58 HC
- Scripted + spontaneous speech
- 3,000 audio files



Dataset 3: English

Mobile Device Voice Recordings at King's College London (MDVR-KCL)

Format:

- 37 speakers, each speaker performs two speech tasks
- *Speaker IDs 00-36 (ID-18 missing in Spontaneous Dialogue)*

Spontaneous Dialogue

Topics: describe where you're from, places to visit in London, sports you play

Read Text

Script: The North Wind and the Sun

Notes:

- Good audio quality, no noise
- Some stretches of silence / ambient noise
- Before the speech starts & in between segments, there is brief confirmation of "can you hear me?" between moderator and the speaker
- For Spontaneous Dialogue, speaker is prompted into making conversation but you can kind of hear the moderator (although very faint, sound is like through headphones)
- I can sometimes difference between speakers with PD and HC speakers
 - Really obvious for ID-13

- 2017
- English
- Severity annotations in addition to diagnosis
- 16 PD, 21 HC
- Read Text + Spontaneous Dialogue
- 73 audio files



Project Direction

- Accuracy vs Interpretability



Self-Supervised Learning (SSL) embeddings

- Black box models
- Generating complex numbers
- Can only achieve interpretability using post-hoc interpretability methods that fit explainable features over them (LIME, SHAP)

Feature Extraction using openSMILE toolkit

- Uses a standard set of 88 features (eGeMAPS)
- Low-level descriptors
 - Some are human interpretable: *jitter, shimmer, harmonics to noise ratio*
 - Some are less interpretable: *MFCCs*

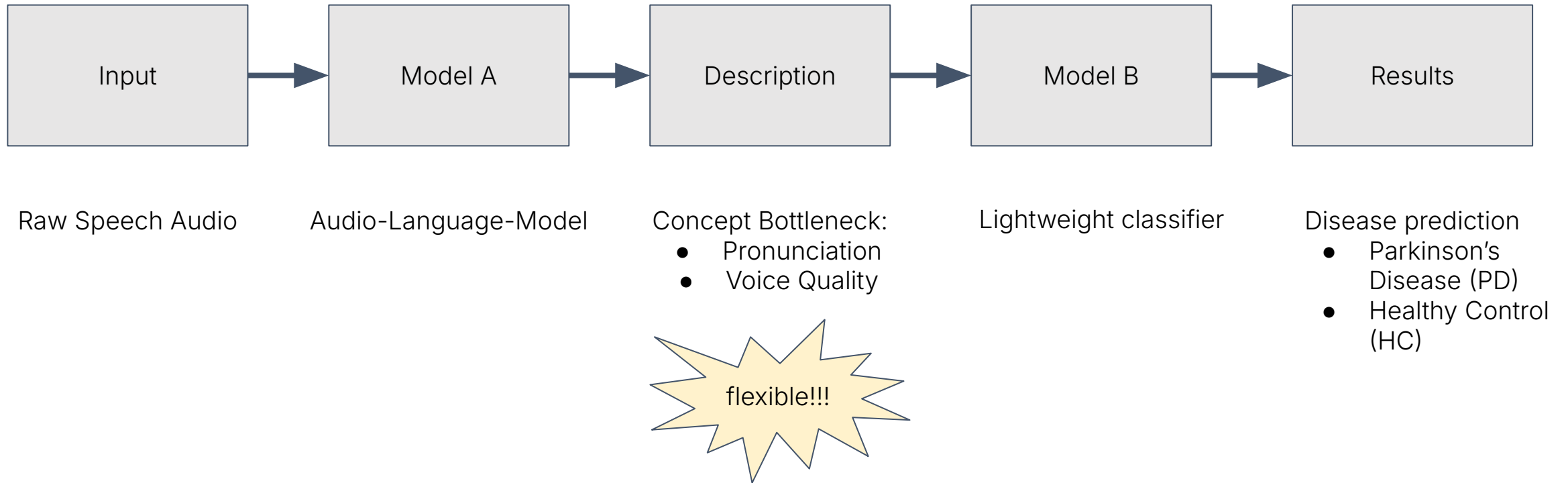
Concept Bottleneck Framework using Audio Language Model (ALM)

- High-level descriptors
 - Clinical Voice Quality: *breathiness, roughness, loudness*
 - Speech Behavior: *slow speech tempo, low intensity, reduced prosodic expression*



Project Direction

Concept Bottleneck Framework



Project Direction

Concept Bottleneck Framework

Research Question: Which descriptors are the most useful to look at?

In terms of clinical interpretability + accurate performance

Clinical Voice Quality

(CVQ): asthenia, breathiness, roughness, strain, loudness, pitch, overall severity

From dataset using CAPE-V and GRBAS scales

- Physiological
- Better bottleneck feature for **depression**

Speech Behavior (SB): slow speech tempo, reduced pitch variability, low intensity, reduced prosodic expression, long or frequent pauses, reduced articulation clarity, vocal strain or tension

- Behavioral
- Better bottleneck feature for **dysarthria**

Pronunciation: Accuracy, Fluency, Prosody

Clinical Features: loudness, H1-H2 log ratio, pitch → take the most important openSMILE features if they're interpretable

Voice Quality: breathiness → for the most important openSMILE features that are *not* interpretable, make them interpretable?

<https://pmc.ncbi.nlm.nih.gov/articles/PMC9613976/>

The accurate differential diagnosis of dysarthria subtypes was based on the quantitative acoustic analysis of 26 speech dimensions related to phonation, articulation, prosody, and timing.



Bridge2AI-Voice Dataset

Updates

- **IRB Consultation** → needs IRB review
 - Personally Identifiable Information
- We currently have access to the version of the dataset that doesn't need IRB approval
 - Sets of features extracted via different toolkits: openSMILE, PRAAT, etc.
 - Can train a speech classifier on these already extracted features
- Start the IRB approval process
 - We would need raw audio if we want to benchmark the feature-based approaches against a voice concept bottleneck FW or SSL embeddings — which require raw audio (?)
 - *Rascal platform*

<https://research.columbia.edu/frequently-asked-questions-hrpoirb>

- ✓ I am analyzing existing data for research. Do I need to submit a protocol to the IRB?

You will need to submit a protocol to the IRB if the analysis of existing data constitutes human subject research.

Under the Department of Health and Human Services (DHHS) regulations “human subject” means a living individual about whom an investigator (whether professional or student) conducting research:

- obtains information or biospecimens through intervention or interaction with living individuals and uses, studies or analyzes the information or biospecimens; or
- obtains, uses, studies, analyzes, or generates identifiable private information or identifiable biospecimens.